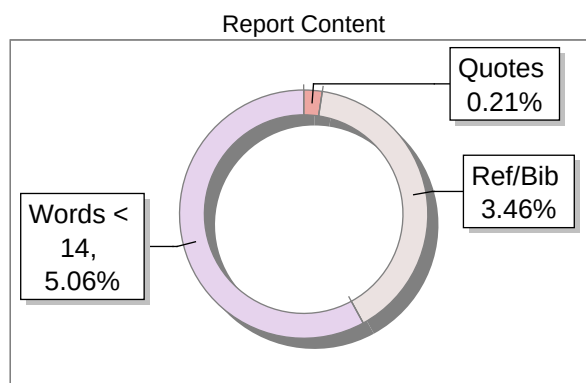
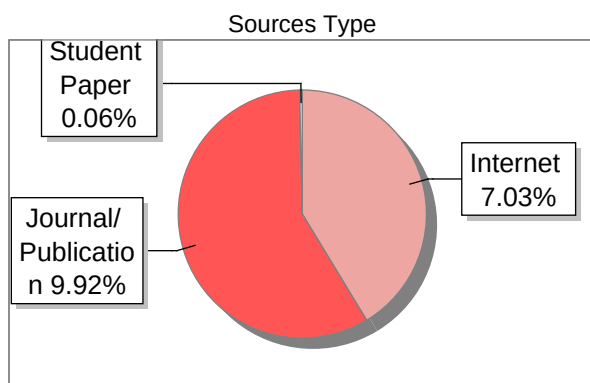


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Enhancing Privacy in Text Data: A Metric-Differential Privacy Approach.

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for the degree of*

MASTER OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING
by

ADEEL AHMAD KHAN CS22M102

Supervisor(s)
Dr.Srinivas Padhmanabhuni



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY TIRUPATI**

May, 2024

DECLARATION

I hereby certify that the ideas expressed in this written submission are entirely mine, and when I have borrowed from others, I have properly credited and referenced the original sources. I further declare that, ⁴⁹to the best of my knowledge, I have not misrepresented, fabricated, or falsified any notion, data, fact, or source in my work and that I have complied with all standards of academic honesty and integrity. I am aware that breaking any of the aforementioned will result in disciplinary action from the Institute and may also result in legal action being taken against the sources who were improperly cited or from whose appropriate authorization was not obtained when required.

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Date: 15-05-2024

Signature
Adeel Ahmad Khan
CS22M102

BONA FIDE CERTIFICATE

This is to certify that the report titled **Enhancing Privacy in Text Data: A Metric-Differential Privacy Approach**, submitted by **Adeel Ahmad Khan**, to the Indian Institute of Technology, Tirupati, for the award of the degree of **Master of Technology**, is a bona fide record of the project work done by him under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Place: Tirupati
Date: 15-05-2024

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ABSTRACT

KEYWORDS: Data;Privacy;Differential Privacy;Metric-DP

With the advancement of Artificial Intelligence, particularly Large Language Models (LLMs), the demand for data to train more effective machine learning models is on the rise. Significant advancements in technology are being fueled by large language models (LLMs), such as Baidu's Wenxin, Google's Bard, and OpenAI's GPT families. It takes access to top-notch datasets covering a variety of conversation genres and styles to train a chatbot LLM that can efficiently follow human direction. With the increase in model size the need for additional training data is also increasing. However, while this surge in data utilisation has tremendous potential, it also has a risk. Sensitive information in the wrong hands can cause significant harm, not only financially and socially, but to the extent of causing harm to an entire nation.

Concerns about user data privacy and its use in training machine learning models have grown exponentially with the expansion of data collection. Over the last ten years, Differential Privacy (DP) has emerged as the predominant norm for Privacy Preserving Data Mining. It is frequently used for a variety of data analytic jobs and offers a strong privacy guarantee. Metric-DP is the most recent development in Differential Privacy-based privacy-preserving text perturbation, providing robust privacy guarantees with low utility compromise ($\leq 2\%$). This method involves adding precisely calibrated noise to the vector representations of words in a high-dimensional space described by word embedding models. It was first designed to achieve geo-indistinguishability in location data. Though several enhancements and studies have been proposed, none have explicitly addressed improving the privacy guarantee. In this study, we have thoroughly surveyed and examined the advantages and disadvantages of a variety of privacy-preserving data mining techniques. We then explore differential privacy in depth and gain a thorough comprehension of the underlying mathematics. Finally, we enhance the method of applying m-DP for text perturbation by introducing an additional step to extend the algorithm

for adding an extra layer of plausible deniability. This improvement enhances privacy assurance at the cost of sacrificing utility. We then compare the performance of this extended technique using GloVe (50d and 300d) embeddings to a baseline model. We apply sentiment analysis on the IMDB Dataset using this algorithm and evaluate the results. The findings imply that the algorithm works effectively, sacrificing relatively little utility/accuracy ($\leq 4\%$) while offering a higher level of privacy guarantee appropriate for gathering and using data in real-world situations.

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ABBREVIATIONS

PPDM	Privacy Preserving Data Mining
SMC	Secure Multi-Party Computation
DP	Differential Privacy
LDP	Local Differential Privacy
GDP	Global Differential Privacy
m-DP	Metric Differential Privacy
ML	Machine Learning
NLP	Natural Language Processing
AI	Artificial Intelligence
HUI	High Utility Itemset
FI	Frequent Itemset
PII	Personally Identifiable Information
QI	Quasi Identifiers

CHAPTER 1

INTRODUCTION

1.1 Motivation

Large Language Models (LLMs) saw a substantial upsurge in interest in 2023. By December 2022, ChatGPT alone had over **57 million** users worldwide, according to Statista. ¹⁰⁰ With **100 million users** active every week, this number climbed to around **120 million** by January 2023 and then to **180.5 million** by April 2024. According to a research by **springapps.com**, LLMs will be used by about **750 million** apps by 2025. The LLM market is expected to experience a substantial growth, with a projected CAGR of 79.80% between 2024 and 2030. The market is anticipated to reach a value of USD **259.82 billion** by 2030, up from USD 1.59 billion in 2023. **Natural language processing** (NLP) has undergone a revolution thanks to the fine-tuning of large language models (LLMs), which now provide unmatched ⁶⁵ **performance in tasks like sentiment analysis, text** production, and language translation. The requirement for data unique to NLP business demands will inevitably rise ³⁰ **as a result of the** need to fine-tune models. According to the proposed scaling rules **J.Kaplan et al. (2020)** and **J.Hoffmann et al. (2022)**, the requirement for more training data rises with model size. But the increasing trend in ⁸⁷ **data** collection raises questions about how private information might be misused if it becomes public. The 2024 Global Threat Intelligence Report from Flashpoint states that over **17 billion personal records** were compromised in 2023, accounting for a **34.5%** increase in reported data breach events. The study "Membership Inference Attacks Against Machine Learning Models" demonstrates the feasibility of inferring the presence of a record in the dataset based on the output of a machine learning model. This highlights the heightened responsibility for safeguarding personal privacy. This report offers a summary of the different industry approaches to ¹ **privacy protection in data mining** scenarios. It proposes an extension to an algorithm for data privatization using Metric-DP and dives further into the mathematics of Differential Privacy. The performance of this expansion is assessed on GloVe word embeddings.

1.2 Objectives :

Objective 1-: Perform a brief analysis of Different Privacy-Preserving Data Mining (PPDM) methods.

Objective 2-: Improve the existing Metric-DP algorithm by adding a new step. Evaluate its performance by applying it to the IMDB dataset, which includes both privatised and non-privatized data for sentiment analysis. Finally, evaluate the accuracy of the model.

Objective 3-: To evaluate and plot the change in accuracy for various values of ϵ and illustrate the trade-off between privacy and utility.

1.3 Report Outline :

The report is structured as follows:

Chapter 2 :Literature Review : This chapter focuses on study of various techniques employed in Privacy Preserving Data Mining(PPDM), exploring their advantages and limitations.

Chapter 3: Background Information: This chapter describes the underlying theory related to the project.

Chapter 4: Experiment and Results: This chapter contains experiment done .

Chapter 5:Summary and Future Direction: This chapter contain the overall summary and things that are to be done going forward.

CHAPTER 2

Literature Review

The acknowledgement of the need for techniques to protect sensitive data within datasets, such as rules that should not be extracted or personal information that can jeopardise privacy, was felt more than two decades ago. Multiple methodologies have been suggested to augment the safeguarding of individuals' privacy since then.

Chris and Don (1996) laid the groundwork for privacy-preserving data mining by addressing the evolving need over time and discussing the potential challenges in ensuring data security. They discussed the security and privacy implications of data mining, highlighting the potential risks and concerns associated with the practice. They introduced the need for methods to protect Privacy in data Mining. They also explored the problems of providing access to certain types of information while restricting access to others, particularly in the context of multilevel secure databases and access control policies. They discussed the need for preventing undesirable mining of published data and suggested various approaches and research directions to address this issue.

Aim of the paper: To discuss the security and privacy implications of data mining, identify potential problems, propose solutions, and ⁵¹outline ideas for further research in this area.

Input: The paper focuses on the input of data mining, specifically the information that is made available for analysis. It raises concerns about the possibility of giving out more information than intended and the need to specify what information users are allowed to obtain from published data.

Output: The paper emphasizes the importance of protecting sensitive information and preventing the discovery of "interesting" rules through data mining. It suggests techniques such as altering or limiting the data, generating false rules, and summarizing the data to prevent undesired mining while still allowing general trends to be determined.

Process: The paper discusses the need to understand mining algorithms to determine how they decide if a rule or output is interesting. This understanding enables the alteration or limitation of data to prevent the discovery of sensitive information.

Limitations: The paper acknowledges the difficulty of knowing when a public version

of a database is not amenable to mining and the challenge of determining the appropriate level of access to data. It suggests the use of auditing and aggregation control techniques but does not provide specific details on their limitations.

G. Jelin and E. (2017) conducted a thorough review, providing an overview of the various approaches and methodologies used in Privacy-Preserving Data Mining (PPDM).

The techniques are as follows -

Perturbation Techniques in Privacy Preserving Data Mining (PPDM):

Perturbation techniques involve modifying data through random processes to protect privacy .

Advantages of perturbation techniques include preserving various attributes independently and providing a level of privacy for individual data .

Limitations of perturbation techniques include the inability to regenerate the original data values and the potential loss of information during the perturbation process .

Perturbation techniques offer a way to protect sensitive data while introducing some level of distortion to maintain privacy in PPDM applications.

Anonymization in PPDM:

Anonymization involves removing identifying information from individual records to protect privacy. It can be achieved through techniques like suppression and generalization, which remove or modify sensitive attributes.

Anonymization techniques, such as the **k-anonymity model**, aim to ensure that each record in a dataset is indistinguishable from at least k-1 other records, making it difficult to identify individuals.

Advantages of anonymization include preserving privacy by removing identifying information and allowing for meaningful analysis of the data. However, it can still be vulnerable to attacks like homogeneity attack and background knowledge attack.

Some of the variants of Anonymization are :

L-closeness

T-diversity

Randomization in PPDM:

Randomization is a frequently used approach in privacy preserving data mining (PPDM) research .

It involves adding noise to the authentic data to create values for each record .

The purpose of randomization is to maintain privacy by making it difficult to recover the original data from the perturbed data .

Advantages of Randomization:

Randomization techniques, such as the Randomized Response scheme and random-noise-based perturbation, can achieve both knowledge discovery and privacy preservation .It has the ability to preserve some semantics and anonymize the entire dataset, providing a balance between utility and privacy.

While randomization is considered a better and efficient process, it still has limitations in terms of information loss and the trade-off between privacy and data utility.

Cryptographic Techniques:

Cryptographic techniques are favored in privacy preserving data mining (PPDM) as they offer accurate data transformation and protection, ensuring better privacy and data utility. The advantages of cryptographic techniques include providing good security for the data, preserving privacy, and maintaining data utility.

Scaling cryptographic techniques can be challenging, especially when multiple parties are involved.

One of the limitations of cryptographic techniques is the difficulty in implementing them in scenarios involving malicious models, as there are fewer studies conducted in this area.

Yousra *et al.* (2015) delved deeply into these techniques, providing detailed explanations of the methods discussed in G. Jelin and E. (2017) and exploring recent advancements in the field.

A summary of the Methods Described is as follows :

Differential Privacy Model:

Differential privacy is a model that aims to provide maximum security to private statistical databases by minimizing the chances of records identification.

It is used to protect sensitive information such as medical records, voter registration information, email usage, and tourism data.

The primary goal is to provide global statistical information about the data publicly available while protecting the privacy of individuals whose information is contained in the dataset.

The concept of "indistinguishability" or "differential privacy" ensures privacy in the context of statistical databases. It involves protecting data at storage and transmis-

sion through data security protocols and taking additional steps to protect individuals' confidentiality in the data.

Its biggest **advantage** is that it provides maximum security to private statistical databases by minimizing the chances of records identification.

Its **limitation** is that the data processed by the differential privacy model may still be susceptible to certain attacks, such as internet phishing.

Data Distortion Dependent Methods:

Data distortion dependent methods are techniques used in privacy preserving data mining to protect sensitive information by introducing controlled perturbations or modifications to the data. These methods aim to strike a balance between preserving privacy and maintaining data utility.

Some of the variants are :

Perturbation: Introducing random noise or perturbations to the data to prevent individual records from being identified.

Blocking: Grouping similar records together to hide individual identities.

Aggregation or Merging: Combining multiple records into a single representation to protect individual identities.

Swapping: Exchanging attribute values between records to further obfuscate identities.

Sampling: Selecting a subset of the data for analysis instead of using the entire dataset, reducing the risk of privacy breaches.

Advantage of Data distortion methods is that they enhance privacy by obscuring individual records, enabling data sharing with reduced privacy risks across various data mining algorithms, including classification, association rule mining, and clustering.

Its **Limitations** are that data loss or distortion are possible, impacting result accuracy and utility. The effectiveness of privacy protection varies based on the technique and dataset characteristics. Despite distortion, advanced techniques or adversarial attacks may still unveil sensitive information from the data.

Association Rules Based Methods:

Association rules based methods are techniques used in privacy preserving data mining to discover interesting relationships or patterns in datasets while preserving the privacy of sensitive information. These methods focus on identifying associations between different items or attributes in a dataset.

Some of its variants are:

Fast Hiding Sensitive Association Rules (FHSAR) algorithm: Introduces a strategy to avoid hidden failures and improve the efficiency of the system.

Genetic Algorithm (GA) based technique: Hides sensitive association rules and enhances the security of the database while maintaining the utility and accuracy of the dataset.

Exact border-based technique: Obtains an optimal solution to hide sensitive frequent itemsets with minimum extension of the original database.

Advantage : Association rules-based methods balance privacy preservation and data utility, enabling the discovery of meaningful patterns while safeguarding sensitive information in datasets.

Limitations: The effectiveness of association rules-based methods is contingent on the chosen algorithm, technique, and dataset characteristics, potentially involving a trade-off between privacy preservation and the accuracy or completeness of discovered associations.

Classification rules based methods :

It refers to ⁹⁰privacy preserving data mining methods that utilize classification algorithms ⁹¹to protect sensitive information while still allowing for accurate classification.

Its Variants are :

K-anonymity: Achieves anonymization by ensuring that each individual in a dataset is indistinguishable from at least k-1 other individuals. However, it has been shown to have limitations in overcoming certain ¹attacks and is susceptible to internet phishing.

FHSAR algorithm: A method that secures sensitive association rules with fewer side effects. It includes strategies to avoid hidden failures and improve system efficiency.

Genetic algorithm (GA): A multi-objective technique based on GA to hide sensitive association rules and enhance database security while maintaining rule utility.

Exact border-based technique: This technique finds an optimal solution to hide sensitive frequent item sets with minimal extension of the original database. It involves formulating the generation of the database extension as a constraint satisfaction problem and partitioning the universe of items to enhance efficiency.

Advantage: The main advantage of classification based PPDM is that it allows for accurate classification while protecting sensitive information.

Limitations: These methods have limitations in terms of vulnerability to certain attacks and scalability issues.

Distributed method based PPDM

It refers to ⁹⁷privacy preserving data mining techniques that involve distributing data and computation across multiple entities or parties. Some key points about distributed method based PPDM:

Distributed PPDM methods utilize techniques such as **secure multi-party computation**, **perturbation**, and **restricted query** ⁷¹to protect privacy while allowing for collaborative data mining.

Secure multi-party computation involves performing computations on distributed data without revealing individual data points, ensuring privacy.

Perturbation techniques introduce controlled noise or distortion to the ⁷²data to protect sensitive information while maintaining data utility.

Advantage: Distributed PPDM methods enable collaborative data mining while preserving privacy.

Limitations: Ensuring data consistency and accuracy across distributed entities can be challenging.

³Charlotte and Frederik (2018) proposed an algorithm for privacy preserving logistic regression training on homomorphically encrypted datasets. It addresses the challenge of performing the training phase for logistic regression in a privacy-preserving manner, which is more challenging than classifying new data.

Aim of the paper: The aim of the paper is to propose a privacy preserving algorithm for training a logistic regression model on a homomorphically encrypted dataset.

Input: The paper uses two real-life datasets for testing the proposed algorithm.

The first dataset is a genomic dataset provided by the iDASH competition of 2017. It consists of 1581 records, each corresponding to a patient, with 103 covariates and a class variable indicating whether or not the patient has cancer.

The second dataset is a financial dataset provided by an undisclosed bank. It is used to predict fraudulent transactions. The dataset consists of 20,000 records with 32 covariates, containing floating-point values between 0 and 1.

Output: For the genomics dataset, the algorithm produced accurate results with an ¹⁰⁴area under the curve (AUC) ranging from 0.9418 to 0.9599, ³comparable to running standard algorithms on plaintext data.

In the financial fraud detection use case, the algorithm performed well, with the AUC value of the ¹⁰⁷proposed method slightly higher than ³the AUC value of the standard "glmfit"

model .

Process:The proposed algorithm uses homomorphic encryption to compute on encrypted data directly, without decryption. The algorithm utilizes a new iterative method, similar to the fixed Hessian method, but with lower multiplicative complexity.

Limitation:The limitations of the algorithm in handling specific types of data or scenarios, such as high-dimensional datasets or imbalanced class distributions, are not addressed.

Proposed change: ³⁹Comparative studies with other existing privacy preserving techniques for logistic regression should be conducted to evaluate the performance and effectiveness of the proposed algorithm.

Noman *et al.* (2013) proposed an algorithm **DiffGen** ⁵ for privacy preserving data publishing in the context of healthcare data which ensures -differential privacy, provides strong privacy guarantees without making assumptions about an adversary's background knowledge. The algorithm generalizes the raw data in a probabilistic way and adds noise to guarantee -differential privacy.

Aim of the paper:To propose an algorithm DiffGen that ensures -differential privacy for the disclosure of healthcare data containing both relational and set-valued data, while retaining essential information for discriminative analysis.

Input:The paper used the MIMIC dataset which is a real-life dataset that contains heterogeneous health data, including both relational data (e.g., demographics) and set-valued data (e.g., diagnostic codes and laboratory tests) for conducting experiments on privacy-preserving data sharing in healthcare.

Output:DiffGen achieved higher classification accuracy compared to DiffP-C4.5, using scoring functions such as information gain (INFOGAIN) and maximum frequency (MAX). (Specifics not given in paper).

Process:The algorithm ensures -differential privacy by probabilistically generalising raw data and incorporating noise, effectively combining classical generalisation techniques with output perturbation, and by performing key steps such as initialising values in D, establishing a privacy budget, and generalising the dataset using specialisations.

Limitations:The utility of the algorithm may degrade when the output domain size is very large, making it potentially inappropriate to generate synthetic data for large health databases.

Proposed change : The proposed algorithm should be extended to handle sequential

data, as it currently does not consider the order of items in set-valued data, which may provide additional useful information and pose re-identification risks.

Mohammad *et al.* (2009) proposed a new multi-objective method for addressing privacy concerns in association rule mining using genetic algorithms, with the goal of securing the database while preserving the utility and certainty of mined rules. It examines the tournament selection method within genetic algorithms in detail and introduces itemset hiding as a technique to prevent the disclosure of sensitive rules, emphasising the modification of transaction datasets to achieve this goal, as well as considerations of critical transactions and items in the original database.

Aim of the paper : To propose a novel method for privacy preserving in association rule mining based on genetic algorithms.

Input: Not given in the paper.

Output: Not given in the paper.

Limitations: The paper does not discuss the potential side effects of the sanitization problem, which could be important in evaluating the effectiveness of the proposed method.

Proposed change : Enhance the preprocessing phase by providing a more detailed explanation of the selection criteria for transactions that support sensitive items, ensuring better performance in terms of sanitization speed and minimizing the number of modifications needed.

Jimmy *et al.* (2023) introduced a framework addressing data privacy challenges in joint-venture industrial collaborations in the Internet of Things (IoT) environment. They combined federated learning and a "pre-large concept" of data-mining techniques to mine high-utility itemsets (HUIs) from multiple datasets without direct data sharing. This framework, tested on real industrial datasets, demonstrates high accuracy compared to conventional data-mining methods while maintaining dataset privacy.

Aim of paper: To propose a privacy-preserving data-mining framework for collaboration in the Internet of Things (IoT) using federated learning.

Input: While the paper does not mention the datasets used here explicitly due to privacy reasons, it mentions the use of a dataset containing five transactions with nine items (a to i) as an example to explain the algorithm.

Output : The paper discusses the change in the number of FIs with the change in standard deviation and the pre-large threshold, but it does not provide specific details of the output

⁴⁶ in terms of the number of FIs.

Process: The framework integrates federated learning and a "pre-large concept" of data-mining techniques, allowing aggregated learning without direct data sharing and reducing scanning into different datasets. The algorithm divides the dataset into private datasets, mines frequent itemsets using a proposed algorithm, and uses a combiner for encrypted result filtering and decryption to obtain final itemsets.

G. and V. (2016) introduced IMR (I-Buckets based Map Reducing) as a privacy-preserving technique in data mining, involving mapping and reducing data to anonymize sensitive attributes. During the mapping phase, patient data is bucketed based on attributes like age, with removal ⁸ of sensitive information such as name, address, and pincode. The required bucket is then anonymized and released during the reduction phase. This technique aims to balance privacy preservation and utility in the data mining process, ensuring protection of sensitive information while closely aligning with the original data.

Aim of Paper: To propose an IMR (I-Buckets based Map Reducing) based Anonymization technique for privacy preservation in data mining.

Input : It uses a synthetic data set, created from the CKD data set, encompasses 10,20,576 records with 20 attributes or ⁵⁹ variables that provide information about each data point and requires 132 MB of storage space.

Output: The confusion matrix shows that in the privacy preserved dataset 330,540 instances were classified as "a" (ckd) and 149,518 instances were classified as "b" (notckd) while in original dataset the confusion matrix shows that 641,064 instances were classified as "a" (ckd) and 379,512 instances were classified as "b" (notckd).

Process : Anonymization technique based on IMR (I-Buckets based Map Reducing) for data mining privacy preservation, involves the mapping of patient data into buckets based on age, removal of sensitive attributes, anonymization in the reduce phase, pattern-based selection of sensitive attributes, adoption of generalisation and suppression-based anonymization, on-demand anonymization for specific data mining tasks, optimisation using the ABC algorithm, and application of the ECLAT.

Limitations : The approach proposed in the paper does not address the issue of probability distribution changes through iteration, which may lead to incorrect results.

Change Proposed: Further research on the impact of different anonymization methods on data mining tasks, taking into account the specific requirements of the application

and the trade-off between privacy and performance.

Alexandre *et al.* (2002) introduced a framework for privately mining association rules from transactions with categorical items, proposing effective randomization operators to limit privacy breaches and deriving formulae for unbiased support estimation. It focuses on finding frequent itemsets, validates the algorithm on real datasets, and generalizes the "randomized response" method. The paper explores the dependence of support recovery on the permitted breach level and data characteristics.

Aim of Paper: To introduce a framework for privately mining association rules from transactions with categorical items.

Input: The paper uses two datasets soccer dataset used in the study is derived from the 1998 World Cup website's clickstream log. It comprises 6,525,879 transactions, each representing a client's session with specific web requests meeting defined criteria and Mailorder dataset consisted of around 2.9 million transactions, 15,836 items, and around 2.62 items per transaction. Each transaction was the set of items purchased in a single mail order.

Output: The evaluation of privacy breaches reveals a 60 percent breach for transactions of 5 minutes or less. For various itemset sizes and transaction lengths, a 50 percent breach level is consistently observed. In football, however, there is a slight "slip" noted for 9 and 10 item transactions. The level of privacy breach is set conservatively at 50 percent.

Process: The process involves the introduction of effective randomization operators, accompanied by formulae for unbiased support estimation and variance. In this approach, each item is replaced with a new item with a defined probability. The analysis explores the correlation between the ability to recover supports, breach levels, and other data characteristics.

Limitation: The paper does not address the issue of vertical partitioning of data, focusing only on horizontally partitioned data.

Proposed Change: Considering vertical partitioning of data and evaluating the trade-off between privacy and utility.

Jaideep *et al.* (2008) presents a privacy-preserving ID3 algorithm for vertically partitioned data among multiple parties, utilizing modified homomorphic encryption and threshold decryption. It discusses implementation details, including the threshold version of Paillier encryption and a secure protocol for maximum finding. The paper explores using the Gini index as an alternative to information gain based on application and

security constraints.

Aim of Paper: To introduce a generalized privacy-preserving variant of the ID3 algorithm for vertically partitioned data distributed over two or more parties,

Input: The researchers utilized two datasets, the car dataset and the weather dataset, sourced from the UCI repository for classification purposes. The weather dataset comprises 4 attributes and the class, totaling 14 transactions. On the other hand, the car dataset has 6 attributes and the class, with a total of 1728 transactions. To create smaller datasets, the car dataset was randomly resampled, resulting in three subsets: car100 (100 percent of the original transactions, 1728 instances), car50 (50 percent of the original transactions, 864 instances), and car25 (25 percent of the original transactions, 432 instances).

Output: On car and weather datasets from the UCI repository, the build time for the distributed ID3 tree and instance classification was evaluated. The results show that the build time scales linearly with transaction count, whereas increasing the number of parties causes a quadratic expansion in time. The size of the tree influences computation time, with smaller trees taking less time to build. Once the tree is built, instance classification is a quick process.

Process : It uses generalised homomorphic public key encryption and threshold decryption. It describes the threshold Paillier encryption and a secure protocol for determining the maximum in a set of numbers. Based on application and security considerations, the paper suggests the use of the Gini index for decision tree computation.

limitations: The paper does not address the potential impact of noise added to the data in the perturbation approach, which is used to mitigate the impact of noise on the data-mining results.

Proposed Change: Increase the encryption speed or decrease the number of encryptions used to improve the performance of the algorithm .

Charu and Philip (2004) introduced a condensation approach for privacy preserving data mining, where data is grouped into condensed units with pre-defined sizes, maintaining statistical information about mean and correlations; anonymized data is reconstructed using eigen-vectors and eigen-values, preserving inter-attribute correlations and enhancing classification accuracy by eliminating anomalies.

Aim of paper: To propose a condensation approach for privacy-preserving data mining.

Input: Ionosphere dataset is a multivariate dataset that contains radar data collected by a

system in Goose Bay, Labrador. The data consists of 351 instances, each described by 34 features, and is used for classification of radar returns from the ionosphere .

Pima-Indian Dataset contains information on 768 women from a population near Phoenix, Arizona, USA. It contains 8 attributes and 1 target variable.

Output: The accuracy of the classifier on the condensed data was comparable to that of the original data (described in graph).

Process: It involves constructing co-variance matrices to preserve statistical characteristics, determining eigenvectors through matrix decomposition, and aims to create anonymized data with similar statistics. The evaluation includes testing the nearest neighbor classifier accuracy with static and dynamic condensation methods, ensuring the preservation of mean values and correlations within each group.

Limitations: The evaluation of the technique is primarily focused on classification accuracy, and there is limited discussion on the impact of the condensation approach on other data mining tasks such as clustering or regression.

Proposed Change: The evaluation of the technique should be expanded to include other data mining tasks such as clustering or regression, to assess the impact of the condensation approach on a wider range of applications.

CHONG *et al.* (1985) introduced a privacy preserving technique called data distortion by probability distribution, involving identifying the original series' density function, generating a series based on this function, and mapping it to replace the original one. This technique ensures individual privacy while maintaining similar statistical properties. The identification process uses goodness-of-fit tests, and various density functions like Poisson, exponential, etc. are considered. The paper also suggests the frequency imposed distortion method for cases where no density function fits the data. The generated distorted series is then used for joint statistical analysis.

Aim of paper: To introduce and discuss the technique of data distortion by probability distribution.

Input: Authors have considered a hypothetical example of the faculty salaries of a business school that has four divisions (Finance, Economics, Management, and Accounting). specifics of data is not mentioned.

Output: The paper compares statistical estimation accuracy of original, point distorted, and probability distorted series. Probability distortion outperforms point distortion in most statistics, except for mean computation in pooled groups. It also shows the dif-

difficulty in compromising probability distorted data, ensuring maximum exposure for statistical analysis while maintaining individual record confidentiality and robust privacy protection.

Process: It involves three steps: identifying the underlying density function of the original series, generating a series of data from the estimated density function, and mapping and replacing the generated series for the original one.

Limitations: Exact data swapping, a form of probability distortion, is not practically feasible, and approximate data swapping is still in an experimental stage with unproven computational efficiency.

Change Proposed: Use of different density functions and compare their performance in terms of accuracy and statistical estimation.

2.1 Related Works

C.Dwork and A.D.Smith (2006) introduced the concept of Differential Privacy (DP) initially developed to protect the privacy of statistical databases, extended it to encompass the privacy-preserving analysis of data. Their work explores literature on disseminating statistical data while maintaining privacy, emphasizing the utility of noise addition for this purpose.

One of the key contributions of DP is the introduction of ϵ -indistinguishability as a novel measure of privacy leakage. By adjusting noise levels according to the sensitivity of the function, this concept ensures privacy preservation, thereby broadening the scope of private data analysis beyond the realm of statistics.

Furthermore, the research extends the application of DP to general functions that map databases to real vectors. By dynamically adjusting noise levels based on the function's sensitivity, privacy can be maintained even when releasing substantial global information, thereby preserving privacy standards.

In general, a randomised algorithm achieves differential privacy if the distributions of its outputs are similar when run on two neighbouring input databases. The concept of similarity is governed by a parameter $\epsilon \geq 0$, which determines the level of privacy assurance. (where $\epsilon = 0$ indicates perfect privacy, and $\epsilon = \infty$ indicates no privacy).

Aim of paper: The aim of the paper is to explore privacy-preserving statistical databases by calibrating noise to sensitivity in private data analysis.

Input:Not Applicable.

Output:Not Applicable.

Process:The paper extends the study to general functions f , demonstrating that privacy can be maintained by adjusting the standard deviation of the noise according to the sensitivity of the function. This calibration reduces the amount of noise required for certain applications. Additionally, the concept of ϵ -indistinguishability is introduced to characterize privacy in terms of transcripts, ensuring that the noise added maintains privacy regardless of the database size.

Limitations:The inherent limitations of input and output perturbation techniques may restrict their applicability in certain scenarios .

Change Proposed:Introduce a mechanism to dynamically adjust the noise level based on the specific data analysis task to balance privacy and data utility effectively.

The notion of differential privacy is expanded upon by [K.Chatzikokolakis and C.Palamidessi \(2013\)](#) to encompass arbitrary metrics.Their research centres on innovative interpretations of privacy in various settings and delves into mechanisms that go beyond query counting. They study how to operationally characterise d-privacy, how to apply it to databases and other domains, and how to create mechanisms that are universally optimal for sum, average, and percentile queries under particular metrics.

The paper presents operational interpretations of privacy guarantees, discusses how d-privacy restricts adversaries' capabilities, and introduces metrics that define privacy guarantees in the context of statistical databases.

Additionally, the study looks at how metrics are used in differential privacy across different services:

Smart Metres: The study takes into account the privacy implications of precise energy consumption measurements from smart metres, going beyond statistical databases. It emphasises how crucial it is to safeguard privacy in situations where frequent and extensive data collection may result in privacy violations.

Location-Based Services: The paper discusses the difficulty of hiding precise locations in location-based services and argues that suitable metrics and hiding functions should be used to reduce privacy risks related to disclosing exact locations.

Fairness in Data Processing: By using d-privacy as a tool, the study seeks to guarantee that similar people get similar results in data processing. It highlights how metrics are used to investigate different concepts, such as justice, throughout different contexts.

Geo-indistinguishability is a formal privacy notion introduced by [M.E. Andrés and C.Palamidessi \(2012\)](#) for location-based systems that use differential privacy. This idea seeks to protect user location data while allowing approximations to be made public.

They provide a method that adds controlled random noise to user locations in order to achieve geo-indistinguishability. Applications for location-based services benefit from this mechanism without sacrificing the quality of the data. The Euclidean metric is used in the paper to expand the idea of differential privacy to location privacy. It also talks about the difficulties in implementing these mechanisms, such as discretization and truncation. Their review of the literature highlights related works like LISA and SpaceTwist, which use techniques like m-unobservability and nearest neighbour querying to address location privacy. A comparative study of the suggested method and current mechanisms shows that it provides better privacy guarantees.

Aim of paper: The aim of the paper is to introduce geo-indistinguishability as a formal notion of privacy for location-based systems, protecting users' exact locations while allowing approximate information release for desired services

Input: Not Applicable.

Output: Not Applicable.

Process: The paper introduces a mechanism for achieving geo-indistinguishability by adding controlled random noise to the user's location, enhancing location-based service (LBS) applications without compromising quality.

Limitations: The paper does not address the potential degradation of privacy due to correlations between multiple data points, which could affect the level of protection in more complex applications.

Change Proposed: To enhance privacy in complex applications, the authors could develop techniques to manage correlations between multiple data points, maintaining the desired level of protection and enabling the composability of the method

[C.Song and Shmatikov \(2018\)](#) demonstrated that, by introducing an auditing technique designed specifically for deep learning models that generate natural-language text, it is possible to determine membership of a specific data from the outcomes of ML models, with the goal of ensuring adherence to data-protection regulations like GDPR. Its specific goal is to audit text-generation models that have been trained on user data in order to identify any unauthorised uses and protect user privacy.

The author's experiments are centred on developing a black-box auditing technique

that can determine whether or not a particular user's data was used to train a model. This approach is shown to be effective in detecting unauthorised data use, as text-generation models are able to memorise training data sequences without compromising test accuracy. These experiments use datasets that contain private information from chats, messages, and comments. The importance of privacy in ML models trained on sensitive data is highlighted by this decision.

Aim of paper: The goal of the paper is to create a new model auditing method that will assist users in determining whether or not a machine learning model was trained on their data.

Input: The Reddit comments dataset (Reddit) from a randomly chosen month of november with the users with at least 150 but no more than 500 posts, for a total of 83,293 users with 247 posts each on average. The Speaker annotated TED talks dataset (SATED), consists of transcripts from TED talks, totaling 2,324 talks with roughly 271K sentences in each language.

Output: On Reddit, test accuracy of word prediction is 20%, similarly on SATED, test perplexity is 10%.

Process: The paper presents a black-box auditing technique that uses very few model queries to determine whether texts from a given user were used to train a text-generation model among thousands of other users. This auditing technique is based on a novel kind of membership inference that uses text-generation models' memorization capabilities and has shown to be very successful in identifying improper uses of personal data.

Limitations: The effectiveness of the auditing algorithm diminishes as the number of users in the text-generation models increases to 10,000. This decrease in performance negatively impacts the algorithm's ability to detect unauthorized uses of personal data. Additionally, the auditing algorithm proves ineffective against differentially private models, as indicated by performance scores near 0.5, which suggest random guessing. This limitation highlights the challenge of effectively auditing such models.

Change Proposed: Implementing a more robust auditing algorithm that can maintain high performance even with a larger number of users in the text-generation models, potentially by optimizing the query selection process to handle larger datasets more efficiently.

O.Feyisetan and T.Diethe (2019) introduced a formal approach for privacy-preserving text perturbation using a variant of Metric Differential Privacy (d_x -privacy). This ap-

proach applies calibrated noise to word vector representations in high-dimensional spaces defined by word embedding models.

It presents a scalable mechanism for text analysis that satisfies d_x -privacy. This mechanism perturbs individual data records by adding noise to word embeddings and projecting back to obtain perturbed words, ensuring plausible deniability to adversaries. The utility of the mechanism is tied to preserving the semantics of input text, leveraging word embedding's geometry for better preservation. The approach demonstrates practical utility with minimal loss in training binary classifiers. Analytical experiments conducted on word embedding models like GloVe and fastText provide insights into distance distributions, influencing privacy guarantees, and plausible deniability statistics for different embeddings. The approach offers tunable privacy guarantees based on the parameter ϵ , providing superior privacy compared to baseline models while maintaining utility across various tasks, such as training binary classifiers.

Aim of paper: The paper aims to present a formal approach for privacy-preserving text perturbation using d_x -privacy, focusing on maintaining user privacy while learning from their data effectively

Input: Paper uses IMDB Movie review Dataset consisting of 50,000 movie reviews sourced from the Internet Movie Database (IMDb). With each review being labeled as positive or negative based on sentiment. ENRON e-mail Dataset containing data from about 150 users, mostly senior management of Enron, organized into folders. The corpus contains a total of about 0.5 million messages. Insurance QA Dataset Containing 12,889 questions and 21,325 answers in Training set, 2,000 questions and 3,354 answers in Validation Set and 2,000 questions and 3,308 answers in Test Set.

Output: In the experiments conducted on the IMDb dataset, we observed the expected trade-off between privacy and utility. As the ϵ value increases (indicating greater privacy loss), the utility scores improve. Conversely, at smaller values of ϵ , we noted lower utility scores. However, this observation varied across the tasks. The approach showcased enhanced privacy and practical utility, with minimal loss in utility ($< 2\%$ for training binary classifiers). Furthermore, privacy audit experiments conducted against baseline models underscored the insufficiency of current methods in thwarting privacy attacks by auditors..

Process: Word perturbations were applied using mechanisms calibrated on GloVe and FastText embeddings. The results indicated that as the average values of N_w increase

(corresponding to higher values of ϵ), the perturbed words become more similar to the original word. The models were trained on the available training sets and evaluated on privatized versions of the test sets, ensuring privacy during evaluation and vice versa.

Limitations: The approach presented in the paper relies on carefully calibrated noise applied to vector representations of words in high-dimensional spaces, which is computationally intensive and costly to implement in real-time systems.

Change Proposed: Enhance the method by introducing an additional step to extend the algorithm for adding an extra layer of plausible deniability.

In order to address domain-specific challenges, [S. Arnold and S. Weinzierl \(2023\)](#) proposed a modification to text-to-text privatisation using Metric Differential Privacy, ⁴¹ as described in [O. Feyisetan and T. Diethe \(2019\)](#). It presents a mechanism for privatisation called MADLIB that adds noise to word embeddings, but it also emphasises the difficulties in controlling noise magnitude variations. In this context, previous studies have focused on geometric properties, in particular, scaling indistinguishability by distance in embedding spaces. By adding candidate selection based on matching grammatical properties, the study improves MADLIB and maintains privacy while improving downstream task performance. By incorporating grammatical features such as part-of-speech tags into the privatisation procedure, syntactic roles in surrogate texts are better preserved than in co-occurrence-based approaches. Furthermore, the study compares and contrasts Metric Differential Privacy with conventional anonymization methods such as k-anonymity highlighting the former's robustness in safeguarding text data privacy.

Aim of paper: The paper aims to enhance text-to-text privatization by considering the preservation of the grammatical category of words during the substitution process

Input: The study utilizes a subset of ⁶⁷ the 1,000 most frequent words from a 100-dimensional GloVe embedding for analysis

Output: The research demonstrates improved downstream task performance by up to 4.66% while maintaining privacy guarantees through enhanced text privatization techniques

Process: Metric Differential Privacy is employed, involving the addition of noise to word representations in embedding space to ensure the preservation of semantic context during substitutions. The study examines the importance of preserving grammatical categories during text privatization and introduces a candidate selection approach to direct substitutions to words with similar syntactic roles. To maintain the grammatical category

of words post-substitution, the research utilizes the multivariate Laplace mechanisms of MADLIB, incorporating constraints into the candidate selection process.

Limitations: Directing substitutions based on matching grammatical categories may lead to additional information leakage, not fully addressed by the modifications proposed in the study.

Change Proposed: Develop strategies to mitigate potential information leakage that may arise from directing substitutions based on matching grammatical categories, ensuring a more comprehensive privacy-preserving mechanism.

CHAPTER 3

Background Information

3.1 About this Chapter

This chapter provides the foundational literature required for understanding the project. It begins by defining key terms, followed by a concise overview of various privacy-preserving data mining techniques. Then it delves into the fundamentals of differential privacy, including relevant mathematical equations and explanations. Finally, it discusses Metric-Differential Privacy, which was used in the experiments to privatise the dataset while clarifying the mathematics and algorithm. It also includes a graph illustrating the taxonomy of privacy-preserving data mining techniques.

3.2 Basic Definitions

Privacy relates to personal information that a person would not wish others to know without authorization, and to a person's right to be free from the attention of others (UN Declaration of Human Rights, 1948).

Data mining is the process of extracting insights from large datasets in order to facilitate the discovery of previously unknown information in databases. The data mining process can be divided into three major stages:

- **Data Preparation:** At this stage, data is collected, cleaned, and organised to ensure its suitability for analysis.
- **Model Construction and Evaluation:** During this phase, models are built and evaluated based on the prepared data, with the goal of uncovering patterns and relationships.
- **Deployment and Maintenance:** Successful models are put into practice, and ongoing maintenance ensures that they remain relevant and effective.

Rapid advances in computer technology have enabled extensive digital data collection and storage, resulting in an increased capacity for personal data accumulation while

raising ethical concerns about privacy, security, and intellectual property rights.

The three major types of privacy violations during Data Mining are :

- **Unauthorized access to data by individuals or groups.** This is effectively prevented through the implementation of access control measures.
- **The exposure of confidential and sensitive data,** Privacy Preserving Data Mining (PPDM) is employed to prevent this.
- **Disclosure of individual's private or sensitive information,** Privacy Preserving Data Mining (PPDM) is employed to prevent this.

Definition: The practice of applying data mining techniques while simultaneously protecting the sensitive rules/information present in the data being analysed is referred to as privacy-preserving data mining.

Goals of PPDM:

- To transform the data to safeguard sensitive business rules/personal information.
- Simultaneously maintaining the utility of the data.

3.2.1 Terminology

Personally Identifiable Information (PII)/Unique Identifiers: It is any data or information that can be used to identify, locate, or contact a specific person. PII is sensitive information that, if handled incorrectly, can jeopardise an individual's privacy and security. e.g Name, address, SSN, and so on.

Quasi-identifiers in Privacy-Preserving Data Mining (PPDM) refer to attributes or combinations of attributes in a dataset that, when combined with auxiliary information, have the potential to re-identify individuals. When cross-referenced with external information sources or data sets, they can be used to link an individual's record in a de-identified dataset to their true identity.

Plausible deniability refers to the ability to deny with credibility that a specific individual's data was included in the dataset used for analysis. This concept is crucial for protecting individual privacy. Differential privacy provides a level of plausible deniability by adding noise to the query results, ensuring that the presence or absence of any individual's data does not significantly impact the outcome.

3.3 Survey of Privacy Preserving Data Mining Techniques

PPDM techniques can be broadly classified into two categories:

- **Secure Multiparty Computation (SMC):** Cryptographic methods, such as homomorphic encryption, enable multiple parties to collaboratively compute functions over private inputs, ensuring data privacy in applications like classification and clustering. However, these methods are often impractical and expensive in real-world scenarios.
- **Data Modification:** Modifying the original dataset, such as changing values, adding noise, or transforming personally identifiable information (PII), creates a tradeoff between privacy and data utility, as methods such as perturbation and k-anonymity demonstrate.

3.4 Data Modification based Methods

3.4.1 Perturbation Based PPDM:

- Data Distortion, Data Randomization, Noise Obfuscation, Amplification and Random Substitutions are Perturbation based PPDM.
- There are two main categories in data perturbation, one based on probability distribution and another one fixed data perturbation.
- A sensitive attribute is perturbed by addition/multiplication of a noise term e , to get a perturbed attribute $Y = X + e$. or $Y = Xe$.
- Effective techniques and do not need information about other records.
- Reconstruction of original data is not possible in real world data.

3.4.2 Condensation Based PPDM:

- Data is reduced into a predefined size of several groups.
- In each group statistical information (mean and correlation) of different dimension is maintained.
- Original data set is converted into a new anonymized data set (using Eigen value

and Eigen vectors of a covariance matrix) which closely matches the original data.

- Information Loss may be huge in this method.

3.4.3 Anonymization Based PPDM:

- This method employs Generalization and Suppression to prevent linking attacks.
- Generalization entails replacing a value with a broader category, while suppression withholds the value from mining.
- In the k-anonymous table, a data set is k-anonymous (k greater than 1) if each record in the data set is not distinct from at least (k-1) other released records within the same data set.
- The model of l-diversity is proposed as an improvement over k anonymization.
- Background knowledge attack, probabilistic inference attack and Complementary Release Attack are few shortcomings of these techniques.

3.4.4 Differential Privacy:

- ¹⁰ Guarantees that, the probability of any transformed data is almost equal to the input data and thus all transformed data are insensitive to any single individual's data.
- There are two approaches in differential privacy: interactive and non-interactive.
- ⁸⁴ Differential privacy offers strong theoretical and mathematically rigorous guarantee.
- Includes accuracy ² of results and improved utility.
- Has scalability issues.

3.5 SMC based Methods

3.5.1 Hiding Association Rule Based PPDM:

- It is mainly used for hiding sensitive rules in frequent item set Mining.
- Here the main aim is to hide patterns of Individuals or Groups.

- By applying proper privacy preservation technique we can hide the association rule e.g. hiding the frequent item sets.
- Modifying few transactions in the transaction database to decrease support or confidence or sensitive rule without side effect can be achieved.
- When item insertion strategy is used, it creates more artificial pattern and if item removing strategy is used then some frequent itemsets becomes infrequent.
- It affects non sensitive rules.

3.5.2 Classification Based PPDM:

- When a data miner analyses a huge number of customer's data to learn classification rules on their data, the sensitive attributes of these customers need to be protected.
- secure multiparty computation (SMC) approach, which uses cryptography for privacy preservation is used.
- Transforming the original data and then constructing a decision tree based on the transformed data set is also used.
- Used in a distributed environment.
- At the same time, data transformation leads to misclassification
- The communication and computation cost may increase with increase in no. of participants.

3.5.3 Cryptography Based PPDM:

- Used when multiple parties with private databases want to jointly run data mining algorithms without disclosing too much information.
- Encryption keys are used for secure sharing.
- Very effective for protecting the privacy.
- It becomes complicated when more than two parties are involved.
- It does not guarantee that the discovery of the final result may not violate the privacy of individual records.

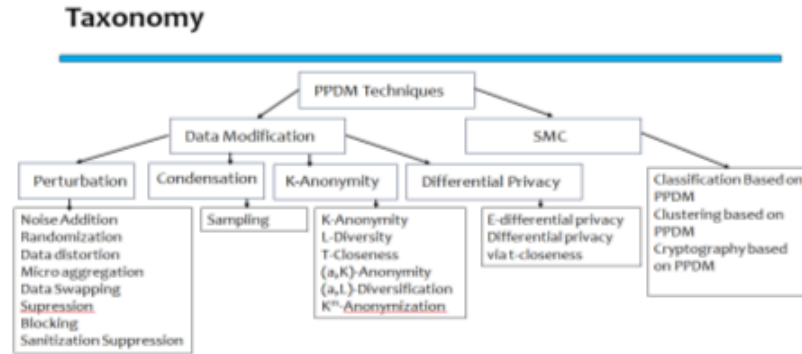


Figure 3.1: Taxonomy of PPDM Techniques

3.6 An introduction to Differential Privacy

Differential Privacy (DP) is a precise definition of privacy that provides solid assurances against potential privacy violations during data processing. It operates without requiring any specific knowledge or computational capabilities from potential adversaries and introduces a clear, measurable, and combinable framework.

Differential Privacy (DP) does not rely on assumptions about a privacy attacker instead, it is founded on the concept of privacy leakage.

The basic idea behind Differential Privacy (DP) according to [Danger \(2022\)](#) is to add carefully calibrated noise to the output of a non-DP algorithm, referred to as a mechanism in this context. This transformation converts it to a DP mechanism. With the goal of maximising the utility of the DP results, the noise calibration is determined by the maximum privacy loss guarantee. To sample the noise, various statistical distributions such as **Laplace**, **Gaussian**, and **Geometrical** are used, resulting in Laplace, Gaussian, and Geometrical DP mechanisms.

When addressing a machine learning task, such as classification, regression, or clustering, **Differential Privacy (DP)** can be incorporated at any stage of the machine learning pipeline.

epsilon ϵ : Epsilon is a parameter in a differentially private algorithm that quantifies the level of privacy protection.

- A smaller epsilon indicates a higher level of privacy. The epsilon value is a measure of the maximum allowable influence of an individual's data on the analysis's outcome.

- In the context of differential privacy, epsilon is commonly interpreted through the concept of "indistinguishability" i.e. With a high probability, the output of a differentially private algorithm should be nearly identical if two datasets differ by only one individual.

Privacy Leakage:

- The amount of information about an individual that can be inferred from the output of a differentially private algorithm is referred to as privacy leakage.
- A lower level of privacy leakage suggests a higher level of privacy protection.
- The goal of differential privacy is to keep personal information private while still providing meaningful and accurate aggregate data.

Privacy Budget :

- It refers to the total amount of privacy that can be expended over multiple uses of a differentially private algorithm.
- Multiple queries or analyses on a dataset may be performed over time in many applications. The privacy budget represents the total amount of privacy that can be consumed across these analyses while still ensuring overall privacy.
- The epsilon parameter is frequently used to control the privacy budget. For example, if you have a privacy budget of one and use epsilon values of 0.1 for two different analyses, the total privacy cost is 0.2, leaving 0.8 for future analyses.

Adjacent Datasets: Two datasets are considered adjacent if one can be obtained from the other by adding, removing, or modifying a single individual's data.

3.6.1 Defining Differential Privacy

Differential privacy is founded on the notion of adjacent datasets. The formal definition of differential privacy is as follows:

Definition 1 (ϵ -DP)

A randomized mechanism or function $M : D \rightarrow R$ with domain D and range R satisfies ϵ -differential privacy if, for any two adjacent inputs d, d_0 and for any subset of

outputs $X \subseteq R$, it holds that:

$$\Pr[M(d) \in X] \leq e^\epsilon \Pr[M(d_0) \in X]$$

The term $\Pr[M(d) \in X]$ denotes the probability that the output of the mechanism M on dataset d belongs to the set X . The equation articulates that the variation in the probability of obtaining a specific outcome using datasets d and d_0 is bounded by the factor e^ϵ .

When dealing with a non-differentially private mechanism, the strategy for introducing differential privacy involves incorporating uncertainty by adding probabilistic noise to the mechanism's output. This addition of noise imparts the essential randomness required to safeguard sensitive user data. As for the choice of the parameter ϵ , it becomes a crucial decision in balancing the trade-off between privacy and utility in the mechanism. An optimal choice for the parameter ϵ should strike a balance, aiming for a low value to enhance privacy while ensuring that a satisfactory level of utility in the responses provided by the differentially private mechanism is maintained.

This definition presupposes that the loss of privacy is solely a consequence of the mechanism's observability. An inherent expansion involves considering the scenario where external and uncontrollable events may also impact the privacy landscape. Hence we arrive at another definition of DP

Definition 2 ((ϵ, δ)-DP)

A randomized mechanism or function $M : D \rightarrow R$ with domain D and range R satisfies (ϵ, δ)-differential privacy if, for any two adjacent inputs $d, d_0 \in D$ and for any subset of outputs $X \subseteq R$, it holds that:

$$\Pr[M(d) \in X] \leq e^\epsilon \Pr[M(d_0) \in X] + \delta$$

Here δ is a variable that signifies the probability of a breach in differential privacy due to factors beyond the control of the differential privacy mechanism.

3.6.2 Features of differential privacy

Composability It asserts that the collective distribution of differentially private mechanisms maintains differential privacy as well. There are two main aspects to consider:

1. **Sequential Composability:** Given n independent mechanisms, denoted as $M_1, \dots, M_n$, each with privacy guarantees of $\epsilon_1, \dots, \epsilon_n$ -differential privacy, respectively, any function $g(M_1, \dots, M_n)$ is $\sum_{i=1}^n \epsilon_i$ -differentially private.
2. **Parallel Composability:** If the aforementioned mechanisms operate on disjoint subsets of the database, then the function g would be $(\max_i \epsilon_i)$ -differentially private.

Robustness: It asserts that any deterministic or randomized function f defined over the image of a (ϵ, δ) -differentially private mechanism M also preserves differential privacy. In other words, $f(M)$ is itself a (ϵ, δ) -differentially private mechanism.

Group Privacy: It introduces a method for safeguarding databases that differ by a count of c rows. This aims to prevent an attacker, armed with auxiliary information, from discerning whether the dataset includes the information of c participants. To achieve this, it is sufficient to constrain the privacy value as ϵc instead of ϵ . In the case of datasets, denoted as d_1 and d_2 , differing on c items, the condition for group privacy is expressed as follows:

$$\Pr[M(d_1) \in X] \leq e^{\epsilon c} \Pr[M(d_2) \in X] + \delta$$

This ensures that each group of c items is ϵ -differentially private protected, while each individual item is ϵc -differentially private protected.

3.6.3 Variance of Differential Privacy

There are more than 200 variants of Differential Privacy but the two most important are:

- **Local Differential privacy.**
- **Global Differential Privacy.**

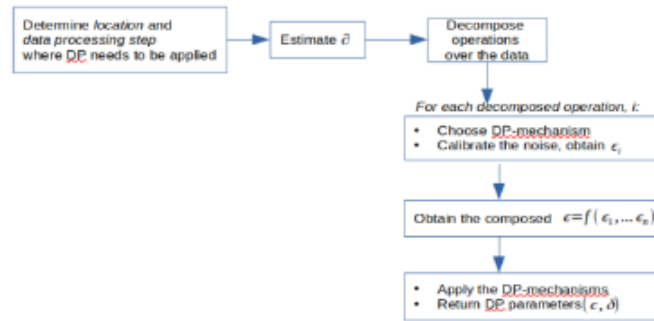


Figure 3.2: A step by step guide on how to incorporate DP in ML

3.6.4 Local Differential Privacy(LDP)

When compared to traditional differential privacy, Local Differential Privacy (LDP) is a privacy assurance mechanism that provides enhanced privacy safeguards for individual users. Because there is no centralised authority in the LDP framework, each user can independently encode and perturb their inputs before transmission. This configuration eliminates the need for users to place their trust in the database maintainer in order to protect their privacy.

LDP extends privacy guarantees to individual users, giving them more control over their data and reducing their reliance on a trusted server. LDP, as a concept that supplements differential privacy, introduces a decentralised approach to privacy protection, making it appropriate for situations in which users may have reservations about trusting the database maintainer.

The growing interest in LDP stems from its effectiveness in enhancing privacy across a wide range of applications, including data analysis and machine learning. This is accomplished by providing enhanced privacy assurances tailored to individual users.

3.6.5 Global Differential Privacy

Global Differential Privacy, distinct from Local Differential Privacy, safeguards individual data in centralized datasets through controlled noise addition, calibrated to the epsilon parameter. This approach facilitates meaningful aggregate analysis, such as computing statistical averages or totals, while acknowledging the inherent trade-off between heightened privacy and potential impact on result accuracy. Relevant across

sectors like healthcare and finance, Global Differential Privacy strikes a balance by enabling centralized analysis, preserving privacy, and ensuring the utility of aggregated results.

3.6.6 Metric Differential Privacy (m-DP)

Metric Differential Privacy (m-DP) is a privacy framework that extends the concept of differential privacy to settings where data is represented in a metric space, such as Euclidean space, and distances between data points are meaningful. m-DP aims to provide a strong privacy guarantee while allowing for meaningful distance-based computations. The key idea behind m-DP is to add noise to the distance function between data points in a way that preserves privacy. The noise is added such that the distance between points in the dataset is not revealing of individual data points. This ensures that even if an attacker knows the distance between two points, they cannot infer much about the actual data points themselves.

A randomized mechanism M satisfies (ϵ, δ) -m-DP if for all datasets D and D' that differ in one element, and for all measurable sets S of metric values:

$$\Pr[M(D) \in S] \leq e^\epsilon \cdot \Pr[M(D') \in S] + \delta$$

where:

- ϵ (epsilon) is the privacy parameter that controls the amount of noise added to the distance function.
- δ (delta) is the probability that the mechanism deviates from the desired privacy guarantee.

The privacy parameter ϵ quantifies the privacy loss due to the mechanism, with smaller values indicating stronger privacy guarantees. The parameter δ is a relaxation parameter that allows for a small probability of failure in satisfying the privacy guarantee.

To achieve MDP, noise is added to the distance function in a way that depends on the sensitivity of the distance function. The sensitivity of the distance function is defined as the maximum change in distance that can occur when a single data point is added or removed from the dataset. The noise added is typically drawn from a distribution that ensures differential privacy, such as the Laplace or Gaussian distribution.

3.6.7 Proposed Mechanism

O.Feyisetan and T.Diethe (2019) address the research challenge of safeguarding private information in text data by leveraging the quantitative privacy guarantees offered by Differential Privacy (DP). Along with these formal privacy criteria, It consider two additional requirements.

- **The privacy mechanism should convert text inputs to text outputs.** ⁹⁸ This allows for seamless integration of the mechanism as a filter into existing text processing pipelines without requiring changes to other system components.
- The second requirement **the ability to scale to handle massive and expanding datasets.**

These specifications force us to find solutions that, like Local Differential Privacy (LDP), process each data record separately. Nevertheless, LDP poses a problem for our application since it demands that, independent of their semantic relationship, every given word x have a non-negligible probability of changing into any other word $\hat{x}$. Because of this limitation, it is practically hard to guarantee that the semantics of x remain intact in the privatised word $\hat{x}$. ⁴³ This is especially true given that the word space expands as the vocabulary of a language grows, and under LDP, the number of words semantically related to x gets progressively smaller.

We turn to d_χ -privacy, a relaxation of local DP originally developed for location privacy, and an extension of m -DP, to address this challenge. This method defines privacy based on the degree of distinguishability between inputs, to overcome the problem of dimensionality in standard LDP and addresses the above-described limitation. With a higher probability assigned to locations close to the current location and a negligible probability assigned to locations in vastly different parts of the planet, this approach allows a mechanism to disclose a user's location in a privacy-preserving manner.

For mechanisms where inputs come from a set X equipped with a distance function $d : X \times X \rightarrow \mathbb{R}^+$ that satisfies the axioms of a metric (i.e., identity of indiscernibles, symmetry, and the triangle inequality), d_χ -privacy is defined mathematically. The particular distance function d that is employed determines the exact definition of d_χ -privacy, which is then parameterized by a privacy parameter $\epsilon > 0$. When, for any $x, x' \in X$, the distributions over outputs of $M(x)$ and $M(x')$ satisfy the following bound, we say that a randomised mechanism $M : X \rightarrow Y$ satisfies ϵd_χ -privacy.

$$\frac{\Pr[M(x) = y]}{\Pr[M(x') = y]} \leq e^{\epsilon d(x, x')}$$

This limit guarantees that the system offers a degree of privacy that rises as the distance between x and x' increases, thereby protecting against the inference of information about individual inputs based on their outputs.

Differential Privacy (DP) is characterised by the probabilistic guarantee expressed in the above equation, which asserts that $\epsilon d(x, x')$ is the bound on the log-likelihood ratio of observing any given output y given two possible inputs x and x' . Local DP differs from d_χ -privacy in that it relates to a particular case of the former, where the distance function is defined as $d(x, x') = 1$ for each x, x' . Nevertheless, there is no way to categorise some pairs of points in X as being closer than others using this Hamming metric. By "remembering almost nothing" about the input, local DP provides very strong privacy by implying a strong notion of indistinguishability of the input. Fortunately, d_χ -privacy is less limiting and permits to scale the indistinguishability of the output distributions by the separation ²⁶ of the corresponding inputs. In particular, the output distributions can be more distinguishable the farther apart a pair of inputs are; this allows the distributions to retain more information about their inputs than in the case of the stricter definition of local DP.

The mechanism follows the "one user, one word" approach and works with individual data records. Our mechanism generates a replacement for the original word x using a d_χ -privacy mechanism $\hat{x} = M(x)$. This replacement is chosen from a carefully crafted probability distribution to ensure that $\hat{x}$ keeps a semantic meaning that is similar to x , ⁴⁵ but at the same time ²⁷ hides any information that might reveal the identity of the user who created x . After perturbing the word in the original record $\hat{x}$, it returns a privatised version of the input word x . The process of achieving this perturbation involves mapping text into a high-dimensional vector space using a predefined word embedding model, adding noise to this vectorial representation, and then projecting back to obtain the perturbed word. From the viewpoint of an adversary observing the perturbed output, the formal privacy guarantees of this mechanism can be understood as offering a degree of plausible deniability V.Bindschaedler and C.A.Gunter (2017) regarding the contents of

x .

The degree to which the output accurately preserves the semantics of the input text is a direct measure of the mechanism's effectiveness. In this sense, our mechanism provides a clear benefit by utilising the geometric properties offered by word embeddings during data perturbation to enable a higher level of semantic preservation. In this work, we measure the preservation of semantics by assessing the privatised data's performance in a downstream machine learning task related to binary sentiment analysis.

Our contributions in this project can be summarized as follows:

- We enhance the application of d_χ privacy for text perturbation discussed in [O.Feyisetan and T.Diethe \(2019\)](#) by introducing an additional step to extend the algorithm, adding an extra layer of plausible deniability.
- We evaluate our model by applying the mechanism to experimental tasks of binary sentiment analysis, at different values of ϵ , demonstrating the trade-off between privacy and utility.

3.6.8 Threat Model

Consider a threat model where a reliable curator gathers words from every user (x) with the goal of giving them to an analyst in an understandable format for use in later tasks (like grouping survey answers or creating machine learning models). The 'one user, one word' model is used to collect data. Presenting words clearly, however, presents the risk of unintentionally providing the analyst with access to data regarding how users are interacting with the system. The analyst and user may have mutual side knowledge, or an ML attack to determine which users commonly use a particular set of words may result in this access. We assume that, in order to solve the downstream tasks of interest, the precise word is not necessary, even though there must be some degree of preservation of the overall semantic meaning.

3.6.9 Algorithm

The mechanism produces a string $\hat{x}$ of the same length after receiving an input string x with $|x|$ words in it. A d_χ -privacy mechanism $M : X \rightarrow X$ is used to privatise x , where $X = W^\ell$ is the space of all strings of length ℓ containing words in a dictionary W . The

metric that is being considered here between strings is obtained using the following word embedding model $\phi : W \rightarrow \mathbb{R}^n$: For any $\ell \geq 1$, given $x, x' \in W^\ell$, we define $d(x, x') = \sum_{i=1}^{\ell} \|\phi(w_i) - \phi(w'_i)\|$, where w_i (resp. w'_i) indicates the i th word of x (resp. x'), and $\|\cdot\|$ indicates the Euclidean norm on $\mathbb{R}^n$. Additionally, we assume that the word embedding is independent of the data that needs to be privatised.

For instance, we could train a new word embedding on an existing dataset, such as GloVe, or use an existing word embedding. The way our mechanism M works is that we first compute the embedding $\phi(w)$ of each word $w \in x$, add appropriately calibrated random noise N to obtain a perturbed embedding $\hat{\phi} = \phi(w) + N$, **find the top 5 nearest neighbours to the perturbed embedding, generate a random number between 1 and 5 (inclusive), choose the nearest neighbour at the index corresponding to the generated random number from the list of nearest neighbours**, and replace w with the word whose embedding is closest to the randomly chosen nearest neighbour. The density of the noise, N , is $p_N(z) \propto \exp(-\varepsilon \|z\|)$, where ε is the mechanism's privacy parameter. The noise is sampled from a n -dimensional distribution. Details of our mechanism's implementation are given in the following pseudo-code:

Algorithm 1: Privacy- Preserving Mechanism

Input: String x containing $|x|$ words, word embedding model ϕ , privacy parameter ε

Output: Perturbed string $\hat{x}$.

- 1 For each word w in x ;
 - 2 Compute embedding $\phi(w)$;
 - 3 Add random noise N to obtain perturbed embedding $\hat{\phi} = \phi(w) + N$;
 - 4 For each perturbed embedding $\hat{\phi}$;
 - 5 Find top 5 nearest neighbors to $\hat{\phi}$;
 - 6 Generate random number r between 1 and 5 (inclusive) ;
 - 7 Choose nearest neighbor at index r as $\hat{w}$;
 - 8 Replace w with $\hat{w}$;
-

Our modification to the algorithm occurs in steps 5 to 8. Instead of selecting the nearest neighbor of the perturbed embedding $\hat{\phi}$, we choose the top 5 nearest neighbors of $\hat{\phi}$, generate a random number between 1 and 5 (inclusive), and select the nearest neighbor corresponding to that random number.

3.6.10 Plausible Deniability Data

To measure ¹⁵ the level of plausible deniability provided by a particular ⁹² selection of the privacy parameter ϵ , we define two metrics. Plausible deniability in text redaction applications refers to the probability of accurately deducing the original input from a sample obtained from the privatisation mechanism. Making sure the original word has a low probability of being released unaltered is necessary to achieve plausible deniability. Furthermore, given an input word, it needs frequently sampled words to show enough variation to make the identity of the input word unclear. A basic difference between d_χ -privacy and Local Differential Privacy (LDP) is that the latter ⁹² requires that almost every possible outcome occurs when a word is perturbed, whereas the former offers a more robust form of plausible deniability. In contrast, d_χ -privacy only requires allocating sufficient probability to words that are similar to the original word in order to stop the output from disclosing the identity of the original word, though it still reveals details about the word's surrounding context. Formally, the metrics we take into consideration are the effective support of the output distribution S_w (i.e., the number of possible output words) for an input w , and the probability $N_w = \Pr[M(w) = w]$ of not changing the input word w . In particular, we define S_w as the size of the smallest set of words that accumulates probability at least $1 - \eta$ for input w , given a small probability parameter $\eta > 0$:

$$S_w = \min \{ |S| : \Pr[M(w) \in S] \geq 1 - \eta \}$$

For (almost) all words in W , **an intuitive setting of ϵ that allows plausible deniability should have a large S_w and a small N_w .**

3.6.11 Affect of our Modification on Plausible Deniability

By modifying the algorithm to select a random neighbor from among the five closest neighbors of the perturbed vector $\hat{\phi}$, we decrease the value of $N_w = \Pr[M(w) = w]$ for each word. This decrease in the probability of $M(w) = w$ increases the plausible deniability provided by the algorithm.

3.6.12 Noise Addition

We start by taking a vector-valued random variable $v = [v_1, \dots, v_n]$ and sampling it from the multivariate normal distribution in order to introduce noise:

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

where the identity matrix is represented by the covariance matrix Σ , the mean μ is centred at the origin, and the dimensionality of the word embedding is denoted by n . Next, we normalise the vector v to make sure it is inside the unit ball. Next, we take a magnitude sample l from the distribution of Gamma.

$$p(x; n, \theta) = \frac{x^{n-1} e^{-x/\theta}}{\Gamma(n) \theta^n}$$

where the embedding dimensionality is n and $\theta = 1/\epsilon$. Next, lv is the output of a noisy vector sample at privacy parameter ϵ . ³⁶ Additional information about this strategy can be found in [X.Wu and Naughton \(2016\)](#).

CHAPTER 4

Experiments and Results

In this section the experiments that were conducted during the semester are listed along with their results. Main aim of these experiments was to fulfill the objectives of this project which were listed in chapter 1 of the report. There are experiments/Implementations that we have done so far and they are as listed below:

1. **Experiment 1:** To implement our algorithm on the IMDB dataset, employing GloVe (50d) for privatization, and then assess its accuracy in a downstream machine learning task focusing on binary sentiment analysis.
2. **Experiment 2:** To implement our algorithm on the IMDB dataset, employing GloVe (300d) for privatization, and then assess its accuracy in a downstream machine learning task focusing on binary sentiment analysis.
3. **Experiment 3:** To create a plot showing the relationship between accuracy and epsilon for both GloVe (50d) and GloVe (300d) embeddings, allowing us to visualize the trade-off between privacy and utility.

4.1 Dataset

We conducted experiments on a textual dataset consisting of IMDB movie reviews for a binary classification task in NLP. To maintain privacy, we anonymized 25,000 reviews from the dataset before using them in our experiments. Table 4.1 provides a summary of the dataset.

Table 4.1: Summary of IMDB dataset

Properties	Values
Training set size	25000
Test set size	25000
Total word count	5,958,157
Vocabulary	79,428

4.2 Embeddings

We conducted experiments using GloVe embeddings of dimensions 50 and 100, pre-trained on the Wikipedia 2014 corpus, in our privacy mechanism to convert original word into a vector and again converting the selected neighbour of perturbed vector back into a word.

Global Vectors for Word Representation, or GloVe for short, is an unsupervised learning algorithm that creates vector representations for words. A corpus's global word-word co-occurrence statistics are analysed as part of the training process. These figures illustrate the frequency with which words occur in the same context throughout the whole corpus.

Revealing linear substructures within word vector space is one of GloVe's primary advantages. As a result, some word relationships, like analogies (for example, "king" is to "queen" as "man" is to "woman"), can be expressed as linear translations inside the vector space. Because of this characteristic, GloVe embeddings are especially helpful in capturing the semantic relationships between words. Further details about the embeddings are given in Table 4.2

Table 4.2: Properties of embeddings

embeddings	GloVe(50d)	GloVe(300d)
Tokens	6 Billions	6 Billions
Vocabulary size	400,000	400,000
cased	uncased	uncased
Dimensions	50	300

4.3 Configuring the Experiments

For each dataset, we demonstrated privacy vs. utility at:

1. **Training time:** We trained the model using perturbed data and evaluated it on unaltered data, simulating a scenario with greater access to private training data. Specifically, the model was trained on a partial training set of 15,000 privatised reviews and tested on the last 10,000 reviews from the test set.

2. **Test time:** We trained the models exclusively on the available training set. However, the evaluation was conducted on a privatized version of the test sets. Specifically, the model was trained on a partial training set of 15,000 non-privatised reviews and tested on the last 10,000 reviews from privatised test set.

For our experiments, we used the Naive Bayes classifier and the evaluation metric was classification accuracy.

For comparison purposes, we employed a Baseline Model that performs sentiment analysis on the IMDB review dataset using a Naive Bayes Classifier. The model was trained on the first 15,000 reviews of the training set and tested on the last 10,000 reviews of the test set. The accuracy obtained was **85.11%**

4.4 Experiment 1:

Aim: To privatize the IMDB dataset using GloVe(50d) embeddings and evaluate the accuracy of the privatized dataset in sentiment classification on both Training time and Test time configurations.

Expected output: The accuracy of the privatized model should be lower than that of the baseline model. The accuracy is expected to increase as the value of ϵ increases, which in turn increases the privacy leakage

4.4.1 Observed Output

We evaluated the model using a dataset privatized with five different values of ϵ , ranging from 10 to 55. The results are as follows:

1. **Training time:** For lowest value of ϵ (i.e 10) the accuracy was **81.15%** while for the highest value of ϵ (i.e 55) the accuracy increased to **81.71%**
2. **Test time:** For lowest value of ϵ (i.e 10) the accuracy was **74.36%** while for the highest value of ϵ (i.e 55) the accuracy increased to **76.34%**

Table 4.3 displays the accuracy values at both Training time and Test time for all five values of ϵ .

Analysis: While the difference in accuracy between the Baseline model and Training time accuracy is approximately 4%, this difference decreases to 3.7% with increasing

Table 4.3: Epsilon values and corresponding accuracy for GloVe(50d) embeddings

ϵ values	Training Time Accuracy	Test Time Accuracy	Baseline accuracy
10	0.812	0.744	0.851
25	0.809	0.752	0.851
35	0.812	0.757	0.851
45	0.816	0.763	0.851
55	0.817	0.764	0.851

values of ϵ . In contrast, the difference between Baseline accuracy and Test time accuracy is 10.8%, decreasing to 8.7% as the value of ϵ increases. This trend confirms that utility increases as privacy guarantee decreases.

4.5 Experiment 2:

Aim: To privatize the IMDB dataset using GloVe(300d) embeddings and evaluate the accuracy of the privatized dataset in sentiment classification on both Training time and Test time configurations.

Expected output: The accuracy of the privatized model should be lower than that of the baseline model. The accuracy is expected to increase as the value of ϵ increases, which in turn increases the privacy leakage.

4.5.1 Observed Output

We evaluated the model using a dataset privatized with five different values of ϵ , ranging from 10 to 55. The results are as follows:

1. **Training time:** For lowest value of ϵ (i.e 10) the accuracy was **61.28%** while for the highest value of ϵ (i.e 55) the accuracy increased to **82.15%**
2. **Test time:** For lowest value of ϵ (i.e 10) the accuracy was **54.32%** while for the highest value of ϵ (i.e 55) the accuracy increased to **77.74%**

Table 4.4 displays the accuracy values at both Training time and Test time for all five values of ϵ .

Analysis: While the difference in accuracy between the Baseline model and Training

Table 4.4: Epsilon values and corresponding accuracy for GloVe(300d) embeddings

ϵ values	Training Time Accuracy	Test Time Accuracy	Baseline accuracy
10	0.613	0.539	0.851
25	0.815	0.713	0.851
35	0.822	0.764	0.851
45	0.812	0.763	0.851
55	0.822	0.777	0.851

time accuracy is huge 23.83%, this difference decreases to 2.96% with increasing values of ϵ . In contrast, the difference between Baseline accuracy and Test time accuracy is 30.79%, decreasing to 7.34% as the value of ϵ increases. This trend confirms that utility increases as privacy guarantee decreases.

4.6 Experiment 3:

To draw the epsilon vs accuracy plot showing the relationship between accuracy and epsilon for both GloVe (50d) and GloVe (300d) embeddings, visualizing the trade-off between privacy and utility and comparing it with the accuracy of the baseline model.

Figure 4.1 shows the variation of Training time accuracy with ϵ for GloVe(50d) embeddings.

Figure 4.2 shows the variation of Test time accuracy with ϵ for GloVe(50d) embeddings.

Figure 4.3 shows the variation of Training time accuracy with ϵ for GloVe(300d) embeddings.

Figure 4.4 shows the variation of Test time accuracy with ϵ for GloVe(300d) embeddings.

4.6.1 Observed Result:

The observations obtained are given as follows:

1. **GloVe (50d):**The minimum and maximum test time accuracies are 0.7436 and 0.7643, respectively. The minimum and maximum training time accuracies are 0.8089 and 0.8171, respectively.

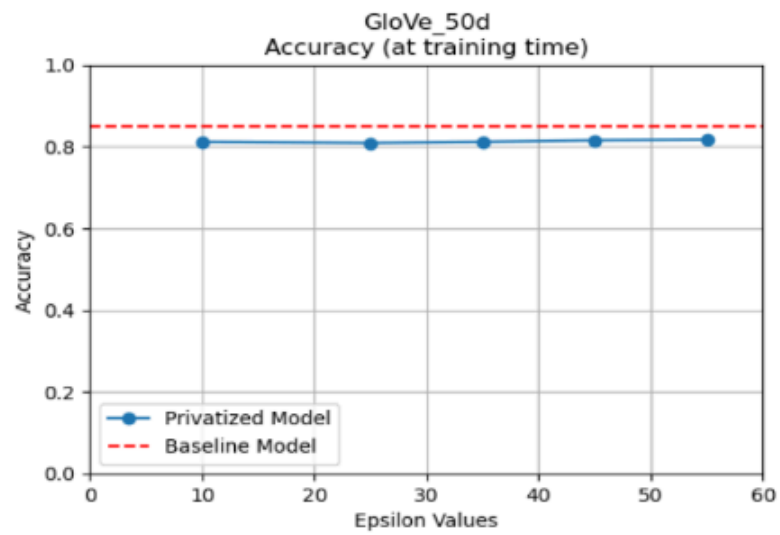


Figure 4.1: Accuracy vs ϵ for Glove(50d) in Training Time

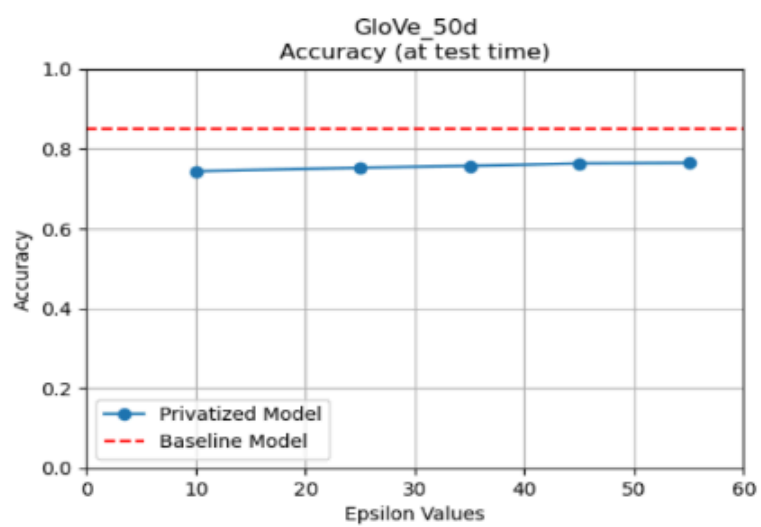


Figure 4.2: Accuracy vs ϵ for Glove(50d) in Test Time

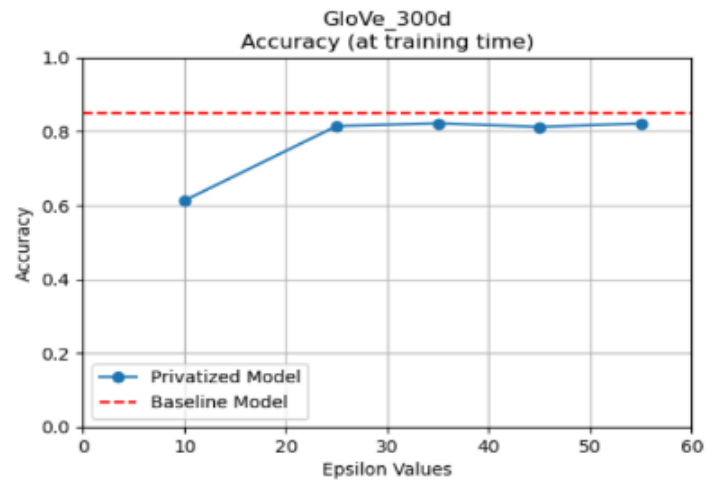


Figure 4.3: Accuracy vs ϵ for Glove(300d) in Training Time

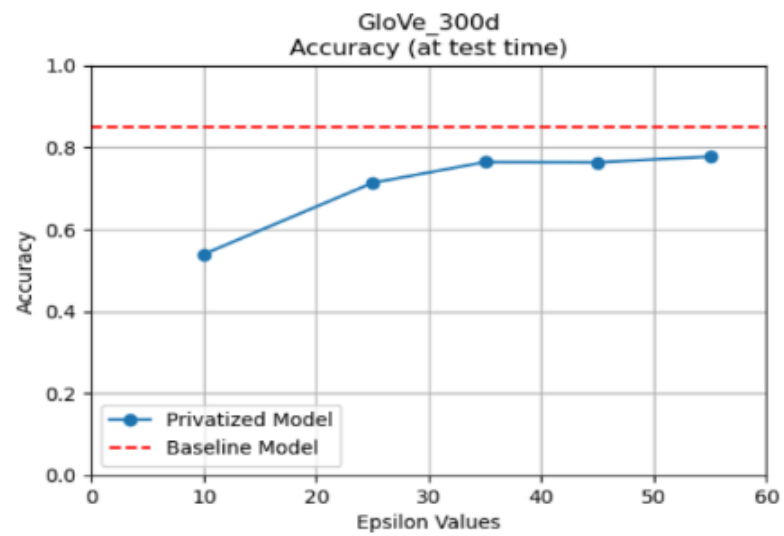


Figure 4.4: Accuracy vs ϵ for Glove(300d) in Test Time

2. **GloVe (300d):**The minimum and maximum test time accuracies are 0.5392 and 0.7774, respectively. The minimum and maximum training time accuracies are 0.6128 and 0.822, respectively.
3. For both GloVe (50d) and GloVe (300d), the test time accuracies are lower than the training time accuracies, indicating that the models perform better when trained on the privatized data as compared to when trained on non-privatized data.
4. The accuracies generally increase with higher epsilon values, indicating that as the privacy guarantee decreases, the utility of the models increases as expected.
5. The accuracies of GloVe (300d) are generally higher than those of GloVe (50d), suggesting that the 300-dimensional embeddings capture more nuanced semantic information than the 50-dimensional embeddings.

Analysis:The proposed algorithm offers a significant level of privacy in the form of plausible deniability, with a trade-off in utility ranging from a minimum of **2.6%** to a maximum of **30.79%**.

4.6.2 Conclusion

From the experiments performed above we can summarize following observations:

1. The proposed algorithm gives an increased privacy at the cost of reduced utility and can be used for privatising information/data.
2. The trade-off between privacy and utility is evident, with higher epsilon values leading to higher accuracies but lower privacy guarantees.
3. GloVe (300d) embeddings tend to outperform GloVe (50d) embeddings in terms of accuracy.

CHAPTER 5

Summary and Future Direction

We investigated a variety of Privacy-Preserving Data Mining (PPDM) Techniques, dissecting their benefits, drawbacks, and various applications. Our investigation included a thorough understanding of Differential Privacy (DP) and its two primary variants, Local DP and Global DP. We explained the notion of differential privacy using metrics and implemented a variant of d-x privacy to privatise our dataset

We implemented and evaluated a privacy mechanism using GloVe embeddings on the IMDB movie review dataset for sentiment analysis. The experiments were designed to assess the trade-off between privacy and utility, with the main objectives being to implement the algorithm using both GloVe (50d) and GloVe (300d) embeddings and visualize the relationship between accuracy and epsilon. The dataset consisted of 25,000 movie reviews, which were anonymized to maintain privacy. The experiments were conducted using a Naive Bayes classifier, with the baseline model achieving an accuracy of 85.11%. Results showed that as epsilon values increased, the accuracy of the models also increased, indicating a trade-off between privacy and utility. GloVe (300d) embeddings generally outperformed GloVe (50d) embeddings in terms of accuracy.

In conclusion, the proposed algorithm offers a good level of privacy with a compromise in utility ($\geq 3\%$ and $\leq 30\%$), making it suitable for privatizing information. The trade-off between privacy and utility was evident, and GloVe (300d) embeddings showed better performance. Visualizations of the relationship between epsilon and accuracy provided a clearer understanding of this trade-off.

Recognizing the critical importance of privacy in Natural Language Processing (NLP), where breaches can have severe consequences for individuals and organizations, this algorithm can be employed to collect data while ensuring user privacy, thereby improving ML model training.

In the future, this work can be extended by evaluating the effect of our modifications on the values of N subscript W to arrive at a more quantified measure of improvement in privacy. Additionally, the performance of this algorithm on other embeddings such as

ELMo, Gilbert, etc., needs to be evaluated to understand the algorithm's suitability and limitations.

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